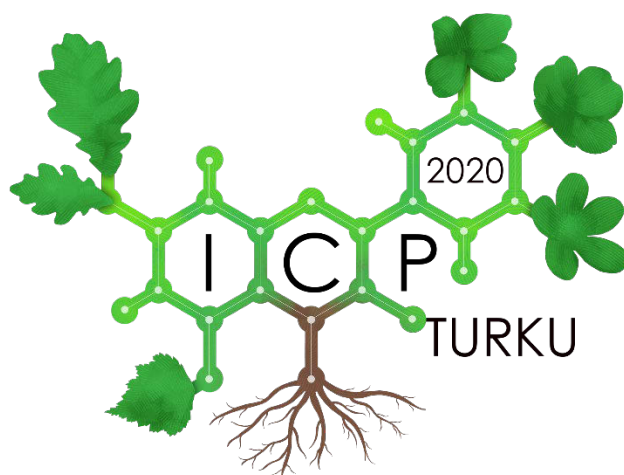




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Surfactant-mediated green extraction of polyphenols from red grape pomace

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MAIN CONCLUSION

The aqueous solutions of surfactants can significantly improve extraction of phenolics from grape pomace regarding to water and therefore non-ionic surfactants could be considered as promising agents for green extraction of polyphenols without usage of organic solvents.

INTRODUCTION

The winery waste is considered an environmental problem as well as low-cost and rich source of high value ingredients, including polyphenols. The waste exploitation and the maximization of polyphenols recovery from grape pomace have become subjects of great interest [1]. A surfactant-based extraction method represents novel approach to the effective solid-liquid extraction of polyphenols from the matrix, following green chemistry principles [2]. The aim of this study was to examine the extraction rate of polyphenols from red grape pomace by the application of selected non-ionic surfactants.

MATERIALS & METHODS

Cabernet Franc red grape pomace was provided by a local winery, dehydrated by lyophilization, grounded and sequentially extracted using aqueous solutions of non-ionic surfactants Brij S10, Brij S20, Brij O10, Brij O20, Tween 20, Tween 60, Tween 80, Tween 85, Triton X-100, Poloxamer 237 and Poloxamer 407. The surfactant concentrations were 0,5%, 1%, 2% and 3%. Water, ethanol and water:ethanol (1:1) were used as control solutions. All solutions were adjusted to $\text{pH } 4.0 \pm 0.05$ and the extraction procedure was performed for 45 minutes with solvent-to-material ratio 100 mL/g at room temperature with constant stirring. The efficiency of the extraction process was determined in terms of total phenolic content, using Folin-Ciocalteu method [3]. The results were expressed as mg of gallic acid equivalents per gram of lyophilized grape pomace (mg GAE/g). Triplicate measurements were taken and results are expressed as mean value $\pm$ standard deviation.

RESULTS & DISCUSSION

Among investigated control solutions, mixture of water and ethanol was the most efficient in the recovery of polyphenols (47.69 ± 2.13). Ethanol and water individually showed lower extraction efficacy, with the total phenolic content of 38.77 ± 1.49 and 25.36 ± 4.37 , respectively.

Poloxamer 407 was more efficient in the recovery of polyphenols than Poloxamer 237 at all investigated concentrations. In comparison to Brij S10 and Brij O10 surfactants, Brij S20 and Brij O20 respectively led to higher yield of target components at all examined concentrations. Brij S20 and Brij O20 showed greater efficacy compared to Triton X-100 as well. Taking into account differences in chemical structures of the tested Poloxamer, Brij and Triton X-100 surfactants, the observed results could indicate that the length of oxyethylene chain as well as the size of hydrophobic part of surfactant molecule significantly affect the extraction efficiency.

The increase in the concentration of surfactants in solutions was followed by the increase in the extraction efficacy for both investigated Poloxamers. Moreover, solutions containing 3% of Poloxamer 407 and Brij S20 exhibited extraction rates comparable to the mixture of water and ethanol, with the total phenolic content of 48.95 ± 3.34 and 44.46 ± 2.96 , respectively. Although

Triton X-100 and Tween surfactants did not reach the yield of polyphenols obtained by the mixture of water and ethanol, they showed greater efficacy than water.

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